



Module 8 Day 2 : Internet Protocols

Activity File: Interpret Protocols

In this exercise, you'll continue to play the role of a security analyst at Acme Corp.

- Several raw network log captures are pulled from a potentially rogue employee, Sally Stealer.
- Your manager tasks you with converting the raw binary data into a readable format and determining the protocol being used.

Instructions

1. View the log files listed below, note that this log file contains multiple log records. Each log record is distinguished by title: Log Record 1, Log Record 2, etc.
2. Convert the binary data of each log record into a readable format with a web tool of your choosing.
3. Document which protocol is used for each log record.
 - Use the following format to document the protocol for each log record: Log Record 1 = SSH.

Bonus

- Interpret the Bonus Log Record to determine which protocol is being used.

Log Files

Log File 1

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0100011101000101010101000010000000101110010000001001000010101000101010001010000001011100110001001011000
11000100001101000010100100100001101110111001101110100001110100010000001110111011010010110010001100111011001
0101110100011100110010111001100011011101101101000011010000101001000011011011101100110011001100101011
0001101110100011010010110111011011000111010001000000110101101100101011000000101101011000010110110
001101001011011001100101000011010000101001010101110000011001110110010011000010110010001100101010101
001001011011100111001101100101011000110111010101100100110010100101101010010011001011100010110101011001
01011100110111010001110011001110100010000000110001000011010000101001010101110011011001010110010001011010
10000010110011101100101011011100111010000111010001000000100110101101110111010011010011000110110001100
00100101111001101010010111000110000001000000010100001011101101001011011100110010001101110110111001110
010000001001110010101000010000000110001001100000010111000110000001110110010000001011101101001011011100011
0110001101000011101100100000011110000011011000110100001010010010000001000001011100000111000011011000110010
1010101110110010101100010010010110100101110100001011110011010100110011001100110011011001100110110001
0000000101000010010110100100001010100010011010100110000101100001000000110110001101001011010110010100100
0000100011101100101011000110110101101110010100100100000010000110110100001110010011011101101011001010
0101110011011000110110001011100011000000101110001100110011001100110101001110010010111000110001001100100110
1110010000001010011011000010110011001100001011100100110100100101110011010100110011011100101110001100110
01101100000110100001010010000010110001101100011011001010111000001110100001110100010000001110100011001010111
10000111010000101111011010000111010001101101011011000010110001100001011100000111000011011000101100011
01100001011101000110100101101110110110001011110111000011010000111010001101101101110000110111100001101
10101101100001011000110000101110000011100000110110001101001011000110110000101101000101101111011011100
01011101111000011011010110110000111011011100010011110100110000000111000111001001011000110100101101011000
010110011101100101001011110110111011001010110001001110000001011000110100101101101100001011001110110010100
101111011000010111000001101110011001110010110000101010001011110010101000111011011100010011110100110000001011
10001110000000110100001010010000010110001101100011011001010111000001110100001011010100010101110011000110
110111101100100011010010110111001100111001110100010000001100111011101001101001110000001011000010000001100
100011001010110011001101100011000010111010001100101000011010000101001000001011000110110010101110000
0111010000101101010011000110000101101110011001110110101110000101100111011001010011101000100000011001010110
111000101101010101010101001100101100011001010111000111011100010011110100110000001011100011100100101100
0110111001100010001110110111000100111101001100000010111000111000

Log File 2

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010001100110100101101100011001010010000001010100011100100110000101101110011100110011001100101110010001
000000101000001110010011011110111010001101111011101101100001000000010100001000110010101000101000
000101001000011101000010100010000000100000001000000011001000110011001100000010000001001100011011110
110011101101001011011100010000001110011011101010110001101100011011001101110011001110101011011
00001011100101110001110010010111000110111000001101000010100010000000100000001000000010000000100000
001000000010000001010010011001010111001101110000011011110110111001110011001010010000001100011011011110110
01000110010100111010001000000101010111001101100101011100100010000001101100011011101100111011001101100101
01100100001000000110100101101110001011000010000001110000011100100110111011000110110010101100100001
0000000101000001100100011001100110000001010010000110100001010001000000010000000100000001000000010
0000001000000010000001010010011100110111000001101111011011100111001100110010100100000011000010111001
00110011100111010001000000100110001101110110011101100101011100010000001110011011010110001101100011011
0010101110011011100110110011001110101011011000100000111001101110101011000110110001101100011011

Log File 3

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0101010001001100010100110110110001100010010110001100100010000001010010011001010110001101011101110010011
001000010000001001100011000010111100101100101011100100011101000100000010110000011100000110100011010
0101100011011000010111010001101001011011101101100010000001000100011000010110100011000010010000001010000
11100100110111011101000110111011000110110111011000111010001000000110100011101000111010001110000001011
0101101110111011001100101011100100010110101110100011011000111001100001101000010100010000000100000001000000
01000000100001101101110110111001100101011011100111010001000000101010001111001011100000110010100111
01000100000010000010111000001110000011011000110100101100011011000010111010001101001011110110111000100000
01000100011000010111010001100001001000000010100000110010001100110010100100001101000010100010000000010000000
10000000100000010101100110010101110010011100110101001011011101100011101000100000010101000100110001010
01100100000001100010010111000110010001000000010100000110000011110000011000000110011001100000011001100101001
0000110100001010001000000010000000100000001001100011001010110111001100111011101000110100000111010001
000000011010100110110000011010000101000100000001000000010000000100000010001010110110011000110110010011110
010111000001110100011001010110010000100000010000001011100000111000001101100011010001101100001011101000
110100101101110110111000100000010001000110000010111010001100001001110100010000001100100001100110110
0110011001100011010000110001001101000011010100110010011001000110000100111001011001010011100101100110011001
101100101011000110011011100110110011000110110001001100101011000100011001100110101011001010011100001100110011
00110011000110011000101100110001101100011010001100010011001100011100000110000011001100011010100110001001100
100011100100110010001101000110000100111110000110100001010

Log File 4

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0100010001101110110110101100001011010010110111000100000010011100110000101101101011001010010000001010011011
1100101110011011101000110010101101101001000000010100001110001011101010110010101110010011100100101001000011
01000010100010000000100000001000000010000001010100011100100110000101101110011100110110000101100011011101000
1101001011011101101110001000000100100101000100001110100010000000110000011110000011000100111000011000100011
01100000110100001010001000000010000000100000001000000100011001101100011000010110011101110011001110100010000
00011000001111000001100000011000100110000001100000010000001010011011101000110000101101110011001000110000101
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00000100000001000000010000000100000001011100010111000101110001000000010111000100000001011100010111000101110
00100000001011100010111000101110001011100010000000101110001011100010111000101110001000000011110100100000010
1001001100101011100110111000011011101101110011100110011001010011101000100000010011010110010101110011011001
10110000101100111011001010010000001101001011100110010000001100001001000000110001011101010110010111001001
11100100001101000010100010000000100000001000000010000000100000001000000010000000101110001100000011
000000110000001000000011000000101110001011100010111000100000001011100010111000101110001000000010111
00010111000101110001011100010000000111101001000000100111101110000011000110111011001000110010100111010001
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000100000001000000010111000101110001011100010111000101110001011100010111000101110001000000010111000
101110001011100010111000100000010111000101110001011100010000000111101001000000101010001110010011101
01011011100110001101100001011101000110010101110010000111010001000000100110101100101011100110110000101
10011101100101001000000110100101110011001000000110111001101110111010000100000011101000111001001110101011011
10011000110110000101110100011001010110010000001101000010100010000000100000001000000010000000100000
001000000010000000101110001011100010111000100000001011100010111000101110001100010010000000101110001
011100010111000101110001000000010111000101110001011100010000000111101001000000101001001100101011000
1101110101011100100111001101101001011011101110001000000110010001100101011100110110010011100100110010101
100100001110100010000001000100011011100100000011100010111010101100101011100100111001001110010011001100
10101100011011101010111001001110011011010010111011001100101011000111100100001101000010100010000000100000
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1011100010111000100000001011100011000000101110001000000010111000101110001011100010111000100000001111
01001000000101101000111010001000000111001001100101011100110110010101110010011100110010101100100001000000
0101000001100000010100100001101000010100010000000100000001000000010000000100000001000000010000000
1011100010111000101110001011100010000000101110001011100010111000100000001011100010111000101110001100

00001000000010111000101110001011100010111000100000001111010010000001001110011011101101110001011010110000101
11010101110100011010000110010101101110011101000110100101100011011000010111010001100101011001000010000001100
1000110000101110100011000010011101000100000010101010110011000110000101100011011001010111000001110100
01100001011000100110110001100101000011010000101000100000001000000010000000100001011101010110010101
110011011101000110100101101110110011100110011101000100000001100010000110100001010001000000010000000100
00000100000001000001011011100110011101101100101011100100010000000100010010010010011100110011101000100000
0011000000001101000010100010000000100000001000000010000001011101010111010001101000011011101110010011
010010111010001111001001000000101001001010010011100110011101000100000001100000001101000010100010000000100
0000010000000100000010000010110010001100100011010010111010001101001011101101110001101110000100000
010100100101001001110011001110100010000000110000000011010000101000100000001000000010000000101000101
11010101100101011100100110100101100101011100110000110100001010001000000010000000100000001000000010
00000010000000100000011000010111000001110000011011000110010101100111011000010111010001100101001011100110001
1011011110110101001110100010000001110100011110010111000001100101001000000100000100101100001000000110001101
0011000011000010111001101110011001000000100100101110000011010000101000100000001000000010000000101
10110101001001100101011100110111000001101110101110011100110011001010010000001001001011100011101000100000
0011011000110010001100110101110100001101000010100000110100001010

Log File 5

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010000010110010001100100011100100110010101110011011100110010000001010010011001010111001101101111011011000111
0101011101000110100101101111011011100010000001010000011100100110111101110100011011110110001101101111011011000
01000000010100001110010011001010111000101110101011001010111001101110100001010010000110100001010001000000010
000000100000001000000100100001100001011100100110010001110111011000010111001001100100100000011101000111100
10111000001100101001110100010000001000101011101000110100001100101011100100110111001100101011101000010000000
1010000011000100101001000011010000101000100000001000000010000001010000011100100110111011101000110
111101100011011011101101100001000000111010001111001011100000110010100111010001000000100100101000001110110
0011010000100000001010000011000001111000001100000011100000110000001100000010100100001101000010100010000000
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001011110100110010100111010001000000011011000001101000010100010000000010000000100000001000000111001
00110111101110100011011101100011011011101101100001000000111001101101001011110100110010100111010001000000011
01000000110100001010001000000001000000010000000100111101110000011000110110111101100100011001010011101
00010000001110010011001010111000101110101011001010111001101110100001000000010100000110001001010010000110100
0010100010000000100000001000000010000001010011011001010111001100100011001010111001000100000010011010100
00010100001100100000011000010110010001100100011100100110010101110011011100110011101000100000010101000110010
1011000110110100001101110011010010110001101111010111100110110001101010011101000110000100111010001
10011001101100010000000101000001110000011100000110100110011000110111001110100110001100111011001110100011011
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01000000101001101100101011011100110010001100101011100100010000001001001010000001000000110000101100100011
001000111001001100101011100110111001100111010001000000011000100110000001011100011000000101110001100000010111
0001100010000110100001010001000000001000000010000000101010001100001011100100110011101100101011101000
010000001001101010000010100001100100000011000010110010001100100011100100110010101110011001110100010
00000011000000110000001110100011000000110000001110100011000000110000001011110011000000110000001110100011000
000110000001110100011000000110000001000000010100000110000001100000011101000110000001110100011000000
11000000111010001100000011000000111010001100000011000000111010001100000011000000101001000011010000101000100
00000100000001000000010000001010100011000010111001001100111011001010111010000100000010010010100000010000
001100001011001000110010001110010011001010111001100111010001000000011000100110000001011100011000000
1011100011000000101110001101100000110100001010

Bonus Log Record

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01001000010000110100100100100000010010000011010000011010000101000100000001000000100000010110110
10001000110100101110010011001010110001101110100011010010110111011011100011101000100000010101010110111001110
01101110000011001010110001101101001011001100110010100101100100010000000101000001100000111100001100110
0110011001100110011001100110011001100110011001100110011000101001010111010001101000010100010000000100000001
0000000100000010010000100001101001001001000000101000001100001011000110110101101100101011101000010000001010
1000111100101110000011001010011101000100000010010000100001101001001000000100001101101110110101101101
011000010110111001100100001000000010100000110000011100000110000001100010010100100001101000010100100100001
000011010010010010000001000011011011101101101101101101101100001011011100110010000100000001011010010000001010
0100110010101100001011001000010000001001100011011101100011011000010110110000100000010100110111010101110000
011100000110111101110010011101000110010101100100001000000100011001100101011000010111010001110101011100100110
01010111001100001101000010100010000000100000001000000010000001000011011011101101101011011011000010110111
0011001000010000001001111011100000110001101101110110010001100101001110100010000001010010011001010110000101
10010000100000010011000110111101100011011000010110110000100000010100110110101110000011100000110111011100
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0000001010000011000010111001001100001011011010110010101110100011001010111001000100000010101000110111011101
0001100001011011000010000001001100011001010110111001100111011010001101000001110100010000000110000000011010
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01010010000001101001011011100010000001100110011100100110000101101101011001010011101000100000001101000101110
10000110100001010001000000010000000100000001000000101101101000011011011101101101011011011000010110111001
10010000101101010100100110010101110011011100000110111011011100111001100101001000000100010001100101011011
000111010001100001001110100010000000110100001011100011000100111000001100010110110111001101011101000011010
0001010